

SHANMUGARAJ

Full-Stack Developer (MERN) | Software Application Development

Chennai, India | +91-8072876474 | shanmugaraj4002@gmail.com | [LinkedIn](#) | [GitHub](#) | [Portfolio](#)

SUMMARY

Full-stack developer and 2025 Electronics & Communication Engineering graduate with hands-on experience building and deploying end-to-end applications using React.js, Node.js, PostgreSQL, and MongoDB. Experienced in designing RESTful APIs, implementing JWT-based authentication and role-based access control (RBAC), and developing scalable backend systems. Strong foundation in data structures, databases, operating systems, and computer networks, with 150+ LeetCode problems solved. Passionate about software development, learning new technologies, and building reliable products that solve real-world problems.

TECHNICAL SKILLS

Languages: JavaScript (ES6+), TypeScript (learning), C++, SQL, Python, HTML5, CSS3

Frontend: React.js, Bootstrap, Responsive UI Design

Backend: Node.js, Express.js, REST API Design, MVC Architecture, Mongoose, Prisma ORM

Databases: PostgreSQL, MongoDB, Query Optimization, Relational Schema Design

Security & Testing: JWT, RBAC, bcrypt, Input Validation, Unit Testing, Integration Testing

DevOps: Git, GitHub, Vercel, Postman, CI/CD (GitHub Actions - learning)

Core CS: Data Structures & Algorithms, OOP, DBMS, Operating Systems, Computer Networks, System Design (learning)

Productivity & Soft Skills: Google Workspace (Docs, Sheets), self-motivated, organizational skills, independent & team collaboration

EDUCATION

B.E., Electronics & Communication Engineering

2021 - 2025 (Graduated)

Jerusalem College of Engineering, Chennai

CGPA: 7.5/10 | No Arrears | Coursework: DSA, OOP, DBMS, Operating Systems, Computer Networks

EXPERIENCE

Software Intern | NSIC - Technical Services Center

2024

- Developed and debugged C/C++ applications for microcontroller platforms, building strong foundations in low-level programming, memory management, and systematic root-cause analysis.
- Integrated AI/ML inference into embedded systems and performed structured testing and validation, applying the same debugging discipline to full-stack project development.

PROJECTS

Tiny-URL Scalable URL Shortener | *Authorization Node.js · Express.js · MongoDB · Redis · REST API · MVC · Caching*

[GitHub](#)

- Built a core URL shortening service with Node.js and Express, generating unique short codes and handling high-volume redirects through RESTful endpoints.
- Integrated Redis caching to serve frequently accessed redirects without hitting the database, cutting average lookup latency.
- Added URL authorization to secure link creation and restrict access to owned links.

Task Flow - Team Task Manager | *React.js, Node.js, Express.js, PostgreSQL, Prisma ORM, JWT, RBAC*

[Live Demo](#) | [GitHub](#)

- Built and deployed a full-stack task management platform with a Kanban board, task filtering, overdue task tracking, and role-based collaboration for team project management.
- Designed MVC RESTful APIs with JWT authentication and admin/member RBAC; optimized PostgreSQL queries using Prisma ORM to improve performance and scalability.
- Kept the codebase clean and production-scalable using modular React components and Git feature branching.

College Transport Tracking System | *Flutter, Dart, Google Maps API, REST API, Real-Time Geolocation*

- Independently learned Flutter/Dart to build a cross-platform live vehicle-tracking app using the Google Maps API; diagnosed and fixed real-time map lag by optimizing REST API polling frequency.

PROBLEM SOLVING

- Solved 150+ LeetCode problems across Arrays, Strings, Trees, Graphs, and Dynamic Programming, with strong command of time and space complexity analysis applied to backend optimization.
- Studying distributed systems and system design fundamentals: scalability, caching, load balancing, database sharding, microservices.

ACHIEVEMENTS

- 2nd Place, College Web Development Competition - built a fully responsive, functional web app recognized among all competing teams.
- Deployed 2 live full-stack applications on Vercel featuring RBAC, JWT authentication, and both SQL and NoSQL databases.